

Since the first recognition in 1839, PV technology has achieved tremendous progress. The solar cell efficiency has increased from 2-3 per cent level in the last century to about 46 per cent in a new generation of PV modules developed in 2010, which is commercially viable. However, there is still much effort needed for efficiency improvement and cost reduction.

Some emerging technologies that can considerably improve energy utilisation efficiency include multi-junction cells, optical frequency shifting, multiple-exciton-generation cells, multiple-energy-level cells, hot-carrier cells, concentration PV systems and hybrid PV systems. However, these technologies are comparatively costly. PVs having quantised band structure, such as quantum wells and quantum dots, can theoretically be as efficient as 60 per cent. In fact, lead-selenide (PbSe) quantum dots have demonstrated such effect in laboratories.

Hybrid solar device

For the single-crystalline single-junction silicon (Si) technology, the conversion efficiency is lower than 30 per cent. Thus a large portion of solar energy is wasted through heat dissipation. Recently, a hybrid solar panel integrating PV, thermoelectric (TE) and hot-water (HW) modules has been developed. With this, the waste heat can be recovered appreciably and electricity generated by using the thermoelectric module, thereby augmenting the overall conversion efficiency of the composite system.

Thermoelectric generator. Thermoelectric generator is a good candidate for thermal energy harvesting. Thermoelectric materials are solid-state energy converters. The combination of thermal, electrical and semiconductor properties allow these to be used to convert waste heat into electricity, or electrical power into cooling and heating directly.

The basic unit of a thermoelectric generator is shown in Fig. 1. It consists of two types of thermoelectric

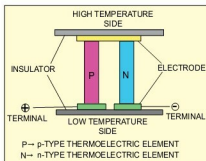


Fig. 1: Basic unit of thermoelectric generator

elements—p-type and n-type semiconductors made of bismuth telluride or silicon germanium. At one end, both the elements are connected together with a high-conductivity metal enabling electrons to flow from n-type to p-type leg and also to have uniform hot-side temperature. On the cold side, two metallic electrodes are connected to the legs separately forming positive and negative terminals.

The thermoelectric generator is electrically separated from the hot and cold sides by a suitable insulator so that there is no short circuit or leakage of electricity. Due to temperature difference, electron-hole pairs are created at the hot end, absorbing heat in the process. The pairs recombine and reject heat at the cold ends. A voltage potential, the Seebeck voltage which drives the hole/electron flow, is created between the hot and cold ends of the thermoelectric element legs.

The Seebeck EMF depends only upon the material and temperature. Figure of merit 'Z' indicates the efficiency of the thermoelectric couple. It can be expressed as $Z = S^2 \sigma / \lambda$, where 'S' is Seebeck coefficient, ' σ ' electrical conductivity and ' λ ' thermal conductivity.

As a solid-state device without any moving part, a thermoelectric generator can be completely silent and extremely reliable. It can be used for years to provide electrical power. The existing thermoelectric modules are expensive and their conversion efficiency is low in non-concentrated solar energy technologies. In order

to improve the performance, the efficiency is to be maximised by selecting the best semiconducting materials with controlled doping. Materials made of heavy atoms like lead and bismuth have low lattice thermal conductivity. Hence compound semiconductors like bismuth telluride (Bi_2Te_3) and lead telluride are useful thermoelectric materials.

Another factor that reduces thermal conductivity is induction of disorderliness of the atomic arrangement in the crystals. In bismuth telluride, if some atom of bismuth is replaced with antimony or some of tellurium is replaced with selenium, the thermal conductivity reduces by a factor of ten, but the electrical properties are retained. As efficiency is inversely proportional to thermal conductivity (λ), it would increase efficiency by a factor of more than two.

Nanotechnology and its potential quantum-scale synthesis lead to new super thermoelectric materials, which give a higher efficiency due to confinement of electrons to two-dimensional quantum wells while suppressing the thermal conductivity. In today's commercial thermoelectric modules, efficiency is about 1.0. But emerging nanotechnology in thermoelectric cells has changed the situation. Recently, researchers working on this concept have developed super thermoelectric materials which can achieve efficiency of about 2.4 for a nanoscale structure (made by alternating layers of thermoelectric materials that both enhance Seebeck coefficient and suppress thermal conductivity at room temperature) with conversion efficiency of about 20 per cent.

Concept of hybrid system

The hybrid system integrates PV, thermoelectric and hot-water modules in a composite structure, which can be installed on the rooftop or in an open ground that receives sunlight. The PV module generates electricity directly from sunlight, whereas the thermoelectric module harvests the solar energy which is otherwise wasted in the PV